

8.3.10

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Question: If the eccentricity of an ellipse is $\frac{5}{8}$ and the distance between its foci is 10, then find the latus rectum of the ellipse.

Solution: Given:

- Eccentricity: $e = \frac{5}{8}$
- Distance between foci: $D = 10$

Step 1: Express the foci for standard ellipse

From B.2.7.3, the foci for standard ellipse are at:

$$\mathbf{F} = \pm \sqrt{\frac{f_0}{\lambda_2(1 - e^2)}} \mathbf{e}_1 \quad (0.1)$$

where f_0 is the scalar term in the conic equation and λ_2 is an eigenvalue of matrix $\mathbf{V}$.

Step 2: Distance between foci

The distance between foci is:

$$D = 2 \sqrt{\frac{f_0}{\lambda_2(1 - e^2)}} \quad (0.2)$$

Step 3: Solve for f_0

From equation (2):

$$\frac{D}{2} = \sqrt{\frac{f_0}{\lambda_2(1 - e^2)}} \quad (0.3)$$

Squaring both sides:

$$\frac{D^2}{4} = \frac{f_0}{\lambda_2(1 - e^2)} \quad (0.4)$$

Therefore:

$$f_0 = \frac{D^2 \lambda_2 (1 - e^2)}{4} \quad (0.5)$$

Step 4: Eigenvalues for standard ellipse

For the standard ellipse, from the conic section theory:

$$\lambda_1 = \frac{\|\mathbf{n}\|^2}{1 - e^2} \quad (0.6)$$

$$\lambda_2 = \|\mathbf{n}\|^2 \quad (0.7)$$

For the standard form with $\|\mathbf{n}\|^2 = 1$:

$$\lambda_1 = \frac{1}{1 - e^2} \quad (0.8)$$

$$\lambda_2 = 1 \quad (0.9)$$

Step 5: Calculate f_0 with $\lambda_2 = 1$

Substituting $\lambda_2 = 1$ into equation (5):

$$f_0 = \frac{D^2(1 - e^2)}{4} \quad (0.10)$$

Therefore:

$$\sqrt{f_0} = \frac{D\sqrt{1 - e^2}}{2} \quad (0.11)$$

Step 6: Matrix formulation for latus rectum The length of latus rectum is given by

:

$$\ell = \frac{2\sqrt{f_0\lambda_1}}{\lambda_2} \quad (0.12)$$

Substituting the values of f_0 , λ_1 and λ_2

$$\ell = \frac{D(1 - e^2)}{e} \quad (0.13)$$

Step 7: Substitute the given values

Given $e = \frac{5}{8}$ and $D = 10$:

$$1 - e^2 = 1 - \left(\frac{5}{8}\right)^2 = 1 - \frac{25}{64} = \frac{39}{64} \quad (0.14)$$

Substituting into the latus rectum formula:

$$\ell = \frac{390 \times 8}{64 \times 5} = \frac{3120}{320} = \frac{39}{4} \quad (0.15)$$

Final Answer: The length of latus rectum is :

$$\boxed{\ell = \frac{39}{4}} \quad (0.16)$$

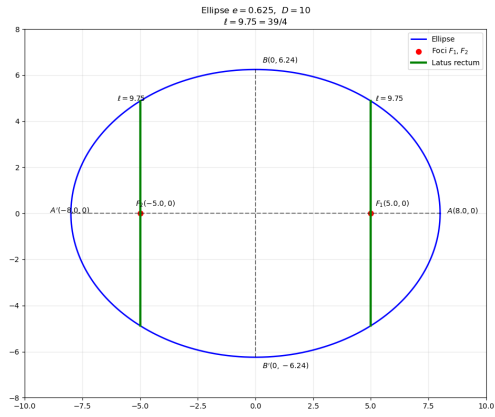


Fig. 0.1: Vector Representation